

SECTION 01

System Overview

ServePoint is a multi-tenant, cloud-ready Hotel Point-of-Sale and Property Management System (PMS-lite) built specifically for the hospitality industry. It provides a unified command center that bridges front-of-house ordering, kitchen operations, room folio management, supplier procurement, and guest loyalty — all from a single interface.

Core Capabilities

 Sales & Order Management

Fast food & beverage ordering from any table or room with multiple payment methods including M-Pesa.

Kitchen Display System

Real-time order ticket management for kitchen staff with live status tracking from Pending to Completed.

Room Folio Management

Charge POS orders directly to a guest's room account. View itemised folios and settle at checkout.

 Inventory & Procurement

Track stock levels in real-time and manage supplier purchase orders through a structured approval workflow.

 Guest Loyalty Program

Accrue and redeem loyalty points for returning guests, with a full transaction audit trail.

 Analytics & Reporting

Revenue trends, top sellers, hourly velocity heatmaps, and payment method breakdowns for management.

Multi-Tenant Architecture

Each hotel operates as a fully isolated tenant. Data, staff accounts, menus, and settings are completely separate — making ServePoint suitable for single properties and hotel chains alike.

SECTION 02

User Roles & Access Control

The system enforces strict **Role-Based Access Control (RBAC)**. There are three operational roles:

ROLE	SCOPE	KEY PERMISSIONS
Super Admin	Platform-level	Manage all tenants, system-wide configuration
Hotel Admin (Manager)	Tenant-level, Full	All modules: Dashboard, POS, Rooms, Procurement, Menu, Analytics, Settings, Staff
Cashier	Tenant-level, Restricted	POS Terminal, Bills, KDS, Guests view, Floor Plan

Note on "Guest"

"Guest" is not a login role. A Guest in ServePoint refers to a hotel patron whose profile is stored in the system for loyalty tracking and room charge functionality. Guests do not have system login access.

SECTION 03

Modules & Feature Breakdown

Dashboard

- **Real-time KPIs:** Today's Revenue, Orders Placed, Average Order Value, Active Tables
- **Sales Velocity Chart:** Hourly revenue breakdown (AM vs PM)
- **Top Sellers:** Ranked by sales volume
- **Recent Orders Feed:** Live order activity with status badges
- **Unread Messages Badge:** Staff communication alerts

POS Terminal

The core order-taking interface used by cashiers and servers. Supports the full checkout flow.

- 1 Browse menu categories or search by item name
- 2 Add items to cart; adjust quantities with +/- controls
- 3 Apply a discount (fixed amount or percentage)
- 4 Select customer (walk-in or loyalty guest)
- 5 Select the room to charge (if posting to a folio)
- 6 Choose payment: **Cash, Card, M-Pesa**, or **Charge to Room**
- 7 Optionally redeem loyalty points as a discount
- 8 Confirm — order, payment, stock deductions, and loyalty updates are all committed atomically

Tax is applied automatically using the tenant's configured rate (e.g., 16% VAT).

Kitchen Display System (KDS)

- Orders appear as ticket cards in real-time
- Status flow: **Pending → Preparing → Ready → Completed**
- Color-coded by elapsed time for urgency awareness
- Completed tickets are archived and linked to order history

Hotel Rooms & Folio Management

Room Statuses: Available | Occupied | Maintenance

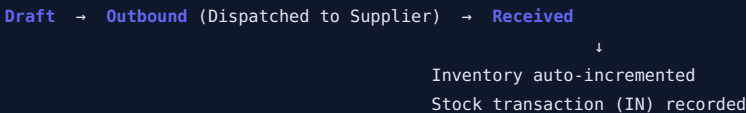
Reservation Flow: Manager creates a booking (Room, Guest name, Dates, Tariff) → Room becomes Occupied. Every POS "Charge to Room" order creates a line in the guest's folio. At checkout, the manager reviews all charges, selects a settlement method, and finalises — the room is released back to Available.

Inventory Management

- Each item tracks: Name, Unit, Current Stock, Low Stock Threshold, Cost Price
- Low-stock items are highlighted with amber alerts
- Every POS sale **automatically deducts stock** based on mapped ingredients
- All movements (IN from procurement, OUT from sales) are logged as auditable stock transactions

Procurement

A structured purchase order workflow from draft to inventory receipt:



Guest & Loyalty Management

- Guest profiles: First/Last Name, Email, Phone, Loyalty Point balance
- **Accrual:** 1 point per KES 100 spent, automatically on every sale
- **Redemption:** Points convert to a cash discount at POS checkout (1 point = KES 1)
- Every accrual and redemption creates an auditable transaction record

Bills & Payments

- Filter by date range, payment method, and order status
- Each entry shows: order number, items, subtotal, tax, discounts, total, method, timestamp
- **Split Bill:** Divide a single order total across multiple payment methods
- Receipts are printable

Floor Plan, Menu Management, Settings

Floor Plan

Drag-and-drop visual restaurant layout. Add, move, and resize tables. Table status updates in real-time and persists across sessions.

Menu Management

Full CRUD for categories and items. Each item can be linked to inventory ingredients with consumption quantities to enable auto stock deduction.

Advanced Analytics

Revenue Trend (30-day), Payment Mix donut, Top Items, Category Performance, Hourly Sales Velocity heatmap.

Settings

Hotel profile, currency, tax rate, receipt footer, staff management with role assignment, and Light/Dark theme toggle.

SECTION 04

Core Business Logic Flows

4.1 Standard Order Lifecycle

```
Cashier selects items + customer + payment method
↓
Cart totalled: subtotal + tax - discount
↓
Single atomic transaction creates:
→ Order + OrderItems
→ Payment record
→ RoomCharge (if method = room_charge)
→ StockTransaction OUT (per ingredient per item)
→ LoyaltyTransaction (if guest selected)
↓
KDS updated in real-time → Bills module updated → Analytics refreshed
```

4.2 Room Charge → Folio → Checkout

```
POS: Order placed as "Charge to Room 101"
↓
RoomCharge record created (Order ↔ Room)
↓
Room 101 card: "TAB: KES 1,856"
↓
Manager opens Room 101 folio → sees itemised order list
↓
Selects payment method → clicks "Finalize Checkout"
↓
All RoomCharges settled → Room status → Available → Folio cleared
```

4.3 Procurement → Inventory Restock

```
Manager: Create PO (Supplier + Line Items: Item, Qty, Price)
↓
PO Status: Draft
↓
Click "Dispatch" → PO Status: Outbound (sent to supplier)
↓
Goods arrive → Click "Commit to Inventory"
↓
PO Status: Received
inventoryItem.stock += quantity (per line item)
StockTransaction (type: IN) logged per item
```

4.4 Loyalty Points — Accrual & Redemption

```
[ACCRUAL] On every POS sale with a guest selected:
points = floor(order_total / 100)
guest.loyaltyPoints += points
LoyaltyTransaction (type: EARN) created

[REDEMPTION] Optional at POS checkout:
Cashier selects "Redeem X points"
Discount applied: 1 point → KES 1
guest.loyaltyPoints -= X
LoyaltyTransaction (type: REDEEM) created
```

SECTION 05

Data Model Summary

Entity	Key Fields
Tenant	name, slug, currency, taxRate, receiptFooter
User	name, email, role, passwordHash, tenantId
Guest	firstName, lastName, email, phone, loyaltyPoints, tenantId
Room	number, type, status, guestName, checkedInAt, tenantId
Booking	roomId, guestName, checkIn, checkOut, status, totalPrice
Category	name, icon, sortOrder, tenantId
MenuItem	name, price, emoji, description, available, categoryId
MenuItemIngredient	menuItemId, inventoryItemId, quantity
InventoryItem	name, unit, currentStock, lowStockThreshold, costPrice
StockTransaction	type (IN/OUT), quantity, inventoryItemId, orderId, notes
Supplier	name, contact, email, phone, tenantId
PurchaseOrder	supplierId, status, totalAmount, orderDate, notes
Order	orderNumber, status, subtotal, tax, discount, total, tableRef
OrderItem	orderId, menuItemId, quantity, unitPrice
Payment	orderId, method, amount
RoomCharge	orderId, roomId, amount, settled
LoyaltyTransaction	guestId, type, points, orderId
Table	name, x, y, width, height, tenantId

SECTION 06

Security Architecture

Layer	Implementation
Authentication	JSON Web Tokens (JWT) — required on every API request as Bearer token
Tenant Isolation	All routes scoped to /api/tenants/:tenantId/ — data strictly isolated per hotel
Role Guards	Middleware enforces permissions — only hotel_admin can access Procurement, Settings, Analytics
Password Security	Bcrypt hashing (salt rounds = 12) — passwords are never stored in plaintext
Auto-Logout	Frontend automatically redirects to login on any 401 Unauthorized response

SECTION 07

Technology Stack

LAYER	TECHNOLOGY
Frontend	React 18, TypeScript, Vite, Recharts, Lucide Icons
Backend	Node.js, Express, TypeScript
Database	PostgreSQL (production) via Prisma ORM
Authentication	JSON Web Tokens (JWT) — stateless, secure
Styling	Vanilla CSS with CSS Custom Properties — full Dark/Light mode
Deployment	Node.js server + static frontend build — compatible with any VPS/cloud host

SECTION 08

Delivery Status — All Modules

Module	Status	Notes
Login & Authentication	✓ Live	Multi-role JWT, auto-logout on expiry
Dashboard	✓ Live	Real-time KPIs + Sales Velocity + Top Sellers
POS Terminal	✓ Live	Full checkout: Cash, Card, M-Pesa, Room Charge
Kitchen Display (KDS)	✓ Live	Real-time ticket management, 4-stage flow
Hotel Rooms & Folio	✓ Live	Booking, room charge, folio, guest checkout
Inventory Management	✓ Live	Auto stock deduction on sales, low-stock alerts
Procurement (PO)	✓ Live	Draft → Outbound → Received + auto restock
Menu Management	✓ Live	CRUD + ingredient mapping
Guest & Loyalty	✓ Live	Points accrual & redemption at POS
Bills & Split Payment	✓ Live	Full audit trail + split bill across methods
Floor Plan	✓ Live	Drag-drop visual layout, persisted
Advanced Analytics	✓ Live	Revenue trend, payment mix, hourly velocity
Settings	✓ Live	Tenant config, tax, staff, theme
Dark Mode (Full System)	✓ Live	All views, modals, and components
Multi-Tenant Isolation	✓ Live	Full data separation per hotel

100% Feature Complete

All 15 system modules are live and operational. The system has been tested end-to-end including the complete order lifecycle, room folio management, procurement restocking, and loyalty point accrual/redemption.

SECTION 09

Demo Access Credentials

The following credentials are pre-loaded in the demo environment with realistic sample data including guests, menu items, inventory, and rooms.

HOTEL ADMIN	manager@lakeside.com / Manager@123
CASHIER	cashier@lakeside.com / Cashier@123
SUPER ADMIN	admin@hotelpos.app / Admin@123
DEMO TENANT	Lakeside Grand Hotel · Currency: KES

